

This Zenodo deposit contains a publicly available description of the Dataset:

"Bulk RNA-seq analysis of the striatum in G2019S LRRK2 knockin mice under a specialized diet".

Dataset Description:

G2019S LRRK2 knockin mice and their wild-type littermates were maintained on a normal chow diet (23.7% kcal from fat) for the first 11 months of life. At that time, animals were randomly assigned to either a high-fat diet group (60% kcal from fat) or a control diet group (10% kcal from fat). Mice remained on their respective diets for an additional five months. At 16 months of age, striatal tissue was collected for bulk RNA-seq analysis. Each experimental group contained biological replicates (n = 4 per group).

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Dataset Name: lee-mouse-bulk-rnaseq-striatum-g2019s-hf-diet, v1.0

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ASAP Project: Are there bidirectional pathological interactions between senescence and LRRK2 models of PD

Project Description: Mutations in LRRK2 gene cause late-onset, autosomal dominant forms of familial PD that are clinically indistinguishable from sporadic PD. In addition, LRRK2 gene polymorphisms are associated with sporadic PD risk. The late disease-onset with these PD-linked genes suggest that interactions with age-related pathways are critical for manifesting or unmasking their pathogenic actions. Studies in experimental models further suggest that PD-linked LRRK2 mutations can induce abnormal mitochondrial function, lysosomal activity, autophagy, proteostasis and inflammatory processes, pathways that are dysregulated by cellular senescence. Our project focuses on determining whether: (1) PD-linked mutations in LRRK2 induce cellular senescence in the rodent brain, (2) increased senescence can exacerbate or unmask PD-like neurodegenerative phenotypes in LRRK2 mouse models, and (3) if removing senescent cells will provide neuroprotection in LRRK2 rodent models of PD.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset